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Section: [R]

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Project Report On

FixHub – On-Demand Home Services Marketplace

Supervised By

DR. MD. IFTEKHARUL MOBIN

Submitted By:

Name	ID
1. Awlad Hossain Abir	24-60107-3
2. Adib Rahman	24-57433-2
3. Sheikh Al Fattah	24-60103-3

Obtained Marks for CO2 and CO3 (Description given in the following page)						
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)	
Evaluation Criteria (CO2)		Total =	Evaluation Criteria (CO3)		Total =	
Requirement fulfillment			Organization of the application			
Validation			Representation and Integration of Database			
Verification			Graphical User Interface			

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall

					reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation of verification processes, and data flow is not considered.	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to ensuring data integrity and appropriate data flow within the system.	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of	Fails to identify and present any	Limited understanding	Lacks depth or relevance to	Integrate the database with	Exhibits an exceptional

Database	understanding or implementation of database. Also failed to integrate the data with the project itself.	of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage .

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Title

FixHub – On-Demand Home Services Marketplace

Introduction

Home repairs and maintenance are often delayed simply because finding a reliable and available service provider is inconvenient. Customers may rely on word-of-mouth recommendations or unverified local contacts, while skilled service providers such as electricians, plumbers, cleaners, and AC technicians may struggle to reach a steady stream of customers. FixHub addresses this gap by acting as a digital marketplace that connects customers with service providers. The platform itself does not perform the services; instead, it manages the booking process and earns revenue through a commission on completed transactions.

FixHub is designed as a desktop application using C# Windows Forms with SQL Server as the backend. The system operates through three user roles: Customer, Admin, and SuperAdmin. It manages the complete lifecycle of a home-service request, from browsing and booking a service to provider assignment, payment, status tracking, and post-service review. At the same time, Admins and the SuperAdmin receive tools to manage service providers, services, complaints, and platform revenue.

CASE STUDY

FixHub is an on-demand home services marketplace management system designed to connect customers directly with individual service providers through a single digital platform. In many real-life situations, customers struggle to find a reliable and available provider for home repairs and maintenance, comparing prices and ratings, and keeping track of their bookings, while skilled providers such as electricians, plumbers, cleaners and AC technicians struggle to reach a steady stream of customers. FixHub solves this by providing a centralized platform where customers can browse verified providers by category, compare price and rating, book directly, and track the service through to completion. The platform is operated by a SuperAdmin who approves new providers, manages the master list of service categories, monitors and resolves customer complaints, and earns a commission from every completed booking.

The system contains three main actors: Customer, Admin, and SuperAdmin. Customers are the buyers who search for a provider by category, price range and rating, view a provider's profile and previous customer reviews, book a service for a chosen date and address, pay through the platform, track the booking status, and rate the provider once the service is completed. Each Admin represents an individual service provider who registers directly on the platform; unlike a shop with a catalogue of products, an Admin account holds one provider's own profile, containing their name, category, price, bio and availability. New Admin accounts require SuperAdmin approval before they can log in. SuperAdmin represents the platform owner and has the highest authority: approving, rejecting or

suspending providers, maintaining the master list of service categories, resolving customer complaints escalated by Customers, and monitoring platform-wide bookings and commission revenue.

Several entities exist within the FixHub system to represent this real-world environment. Customers, Admins and SuperAdmins are each stored in their own table, since a provider's registration and approval workflow differs entirely from a customer's account. Every Admin is uniquely identified by an AdminId and belongs to one service category, alongside a price, bio, availability flag, approval status and suspension status. A Booking is uniquely identified by a BookingId and links one Customer to one Admin, along with the scheduled date, address and a status that moves through Pending, Accepted, In Progress and Completed, or Cancelled and Declined. Each paid Booking has one related Payment record, storing the amount paid, the platform's commission, and the payment method. Customers can leave one Review per completed Booking, containing a rating from 1 to 5 and a written comment. Customers can also file a Complaint against a specific Booking, which the SuperAdmin investigates and resolves. A Coupon can optionally be applied to a Booking at checkout to reduce its total.

Money flows from the Customer to the provider (Admin) through the FixHub platform when a Booking is paid for. The system records the payment amount, the commission amount, and the payment status on the associated Payment record. FixHub operates as the intermediary platform and deducts a percentage of each transaction as its commission; the remaining amount represents the provider's earnings. FixHub therefore benefits financially from connecting customers directly with individual service providers, while giving both sides a single, organized booking platform.

Functionalities

(a) Functional requirement list

A. Customer Workflow

Sign Up — Customer registers with name, email, phone, address and password. FixHub validates the required fields and creates a Customer account.

Sign In — Customer logs in with email and password; FixHub verifies the credentials against the Customers table and opens the Customer Dashboard.

Browse Providers — Customer views approved, available providers grouped by service category, with each provider's name, price and average rating shown.

Search and Filter Providers — Customer can filter results by category, price range and minimum rating, and search providers by name; FixHub applies every selected filter together.

View Provider Profile — Customer opens a provider's profile to see their bio, category, price and existing customer reviews before booking.

Book and Pay — Customer selects a provider, confirms the scheduled date, service address and payment method (Pay Now or Pay After), and can apply a coupon code before confirming the booking.

Track Booking Status — Customer opens My Bookings and follows a booking through Pending, Accepted, In Progress and Completed, or Cancelled and Declined.

Cancel Booking — Customer can cancel an eligible booking from the booking details screen.

Leave a Review — Once a booking reaches Completed, the Customer can rate the provider from 1 to 5 stars and write a comment.

File a Complaint — Customer can submit a complaint tied to a specific booking for the SuperAdmin to investigate.

Account Management — Customer can update their profile information and change their password.

B. Admin (Provider) Workflow

Sign Up — Admin registers a provider account with name, email, phone, category, price and bio. The account is created as pending until it is approved.

Sign In — Admin logs in with email and password; FixHub blocks the login with a pending-approval message until the account has been approved by the SuperAdmin.

Manage Profile — Admin updates their bio, category, price and availability toggle; an unavailable provider no longer appears in customer search results.

View Incoming Requests — Admin views incoming bookings with the customer, address, scheduled date and current status.

Accept, Decline or Update a Booking — Admin accepts or declines a pending booking, and updates its status as the job progresses through to Completed.

View Earnings — Admin's dashboard shows the total completed bookings and net earnings after the platform's commission is deducted.

Account Management — Admin can update their profile information and change their password.

C. SuperAdmin Workflow

Sign In — SuperAdmin logs in through a dedicated login form and is routed to a platform-wide dashboard.

Approve, Reject or Suspend Providers — SuperAdmin reviews newly signed-up Admin accounts and approves them, rejects an unapproved account, or suspends and later reactivates an existing provider.

Manage Service Categories — SuperAdmin adds, renames or deletes categories in the master service-category list used throughout the platform.

Monitor and Resolve Complaints — SuperAdmin reviews open customer complaints and records a resolution, optionally suspending the provider involved.

View Platform Dashboard — SuperAdmin dashboard shows providers pending approval, open complaints, active provider count and total commission earned.

View Commission Report — SuperAdmin views total commission earned broken down by service category, calculated from completed bookings and their payments.

(b) User Stories

Customer

I. Customer Registration

As a Customer, I can create a FixHub account, so that I can book home services through the platform.

Details: The Customer selects "Sign Up" and enters their name, email, phone number, address, password and confirm password. Required fields cannot be empty and both passwords must match. After successful validation, a new record is inserted into the Customers table with a securely hashed password, allowing the Customer to log in.

II. Customer Login

As a Customer, I can log in to FixHub, so that I can access my dashboard and manage my bookings.

Details: The Customer enters their email and password on the Login form and selects the Customer role. FixHub checks the Customers table and verifies the password hash. If the

credentials are correct, the Customer Dashboard opens; otherwise an error message is shown.

III. Browse Providers

As a Customer, I can browse available service providers, so that I can find someone who matches my needs.

Details: The Customer opens Browse Providers, where FixHub lists approved and available providers with their category, price and average rating, calculated from the reviews left on their completed bookings.

IV. Search and Filter Providers

As a Customer, I can search and filter providers, so that I can narrow the results to what I actually need.

Details: The Customer can filter the provider list by service category, a price range and a minimum rating, and can search by provider name. FixHub applies all selected filters together and updates the results immediately.

V. View Provider Profile

As a Customer, I can view a provider's profile, so that I can decide whether to book them.

Details: The Customer selects a provider and opens their profile, which shows the provider's bio, category, price and previous customer reviews retrieved from the Reviews table.

VI. Book a Service

As a Customer, I can book a service with a chosen provider, so that the provider can carry out the work I need.

Details: The Customer confirms the service address and preferred date on the Book and Pay form. FixHub validates the required information and creates a new record in the Bookings table linking the Customer and the selected Admin, with a status of Pending.

VII. Apply a Coupon and Pay

As a Customer, I can apply a coupon and choose a payment method, so that I can complete my booking at the correct price.

Details: On the Book and Pay form, the Customer can enter a coupon code, which FixHub validates against the Coupons table for its active status, expiry date and usage limit before applying the discount. The Customer then selects a payment method, and FixHub creates the Booking together with a corresponding Payments record.

VIII. Track Booking Status

As a Customer, I can track the status of my bookings, so that I know how each request is progressing.

Details: The Customer opens My Bookings and sees each booking's provider, scheduled date and current status. FixHub retrieves only bookings belonging to the logged-in Customer, and the status updates as the Admin accepts, progresses or completes the job.

IX. Cancel Booking

As a Customer, I can cancel an eligible booking, so that I do not receive a service I no longer need.

Details: The Customer opens a booking's details and selects Cancel. FixHub updates the Bookings record's status to Cancelled while it is still in a cancellable state.

X. Leave a Review

As a Customer, I can rate and review a completed booking, so that I can share my experience with future customers.

Details: Once a booking reaches Completed, the Customer can open it and submit a rating from 1 to 5 along with a written comment. FixHub inserts the review into the Reviews table, linked to that Booking, Customer and Admin.

XI. File a Complaint

As a Customer, I can file a complaint about a booking, so that FixHub can investigate and resolve the problem.

Details: The Customer opens the booking details and selects File a Complaint, entering a description of the issue. FixHub inserts a new record into the Complaints table with an Open status for the SuperAdmin to review.

XII. Account Management

As a Customer, I can manage my account, so that my information stays accurate and secure.

Details: The Customer opens Account Settings to update their profile details or change their password. FixHub validates the new password before updating the stored hash in the Customers table.

Admin

I. Provider Registration

As an Admin, I can register a provider account, so that I can offer my services on FixHub.

Details: The Admin selects Sign Up and enters their name, email, phone, service category, price and bio. FixHub inserts a new record into the Admins table with IsApproved set to false, so the account is created but cannot log in until it is reviewed.

II. Admin Login

As an Admin, I can log in to FixHub, so that I can manage my bookings and profile.

Details: The Admin enters their email and password and selects the Admin role. FixHub checks the Admins table and blocks the login with a pending-approval message if IsApproved is still false; otherwise the Admin Dashboard opens.

III. Manage Profile

As an Admin, I can manage my provider profile, so that customers see accurate information about my services.

Details: The Admin opens their profile screen and updates their bio, category, price and availability toggle. FixHub updates the corresponding Admins record, and an unavailable provider no longer appears in customer search results.

IV. View Incoming Requests

As an Admin, I can view incoming booking requests, so that I know what work is waiting for me.

Details: The Admin opens Requests and sees each pending booking's customer, address, scheduled date and current status, retrieved from the Bookings table filtered to their own AdminId.

V. Accept, Decline or Update a Booking

As an Admin, I can accept, decline or update the status of a booking, so that customers know the progress of their request.

Details: The Admin selects a booking and chooses Accept, Decline, or updates its status through In Progress and finally Completed. FixHub validates the change and updates the Bookings table, and the new status becomes visible to the Customer immediately.

VI. View Earnings

As an Admin, I can view my earnings, so that I can track how much I have made through FixHub.

Details: The Admin Dashboard displays the total number of completed bookings and the net amount earned after FixHub's commission is deducted, calculated from the Payments table.

VII. Account Management

As an Admin, I can manage my account, so that my login details remain accurate and secure.

Details: The Admin opens Account Settings to update their contact details or change their password. FixHub validates the new password before updating the stored hash in the Admins table.

SuperAdmin

I. SuperAdmin Login

As a SuperAdmin, I can log in to FixHub, so that I can manage and monitor the entire platform.

Details: The SuperAdmin logs in through a dedicated SuperAdmin Login form. FixHub verifies the credentials against the SuperAdmins table before opening the SuperAdmin Dashboard.

II. Approve, Reject and Suspend Providers

As a SuperAdmin, I can approve, reject or suspend provider accounts, so that only trustworthy providers operate on FixHub.

Details: The SuperAdmin opens Manage Admins and sees each provider's approval and suspension status, along with their completed job count and rating. The SuperAdmin can approve a pending provider, reject an unapproved account, or suspend and later reactivate an existing provider, all updating the Admins table.

III. Manage Service Categories

As a SuperAdmin, I can add, rename or delete service categories, so that FixHub maintains an organized master category list.

Details: The SuperAdmin opens Manage Categories and can add a new category, rename an existing one, or delete one that is no longer needed. FixHub inserts, updates or deletes the corresponding record in the ServiceCategories table.

IV. Monitor and Resolve Complaints

As a SuperAdmin, I can monitor and resolve customer complaints, so that service problems are properly investigated.

Details: The SuperAdmin opens Manage Complaints and sees each open complaint's customer, provider and description. After investigating, the SuperAdmin records a resolution and can suspend the provider involved if necessary; FixHub updates the Complaints table and, where applicable, the Admins table.

V. View Platform Dashboard

As a SuperAdmin, I can view the platform dashboard, so that I can monitor FixHub's overall status at a glance.

Details: The SuperAdmin Dashboard shows the number of providers pending approval, open complaints, active providers, and total commission earned, calculated from the Admins, Complaints and Payments tables.

VI. View Commission Report

As a SuperAdmin, I can view the commission earned by category, so that I can monitor FixHub's income across different service types.

Details: The SuperAdmin opens Reports, where FixHub joins completed Bookings with their Payments and the provider's service category, groups the results by category, and totals the commission earned for each one.

ER Diagram

The following Entity-Relationship diagram represents FixHub's data model using Chen notation. Rectangles represent entities, ellipses represent their attributes (with primary keys underlined), and diamonds represent the relationships that connect entities, labeled with their cardinality (1 or M). The Coupons entity is shown as standalone since, in the current schema, coupon usage is validated and its usage count incremented at booking time, but no foreign key is stored on the Bookings table linking a booking to the coupon that was applied to it.

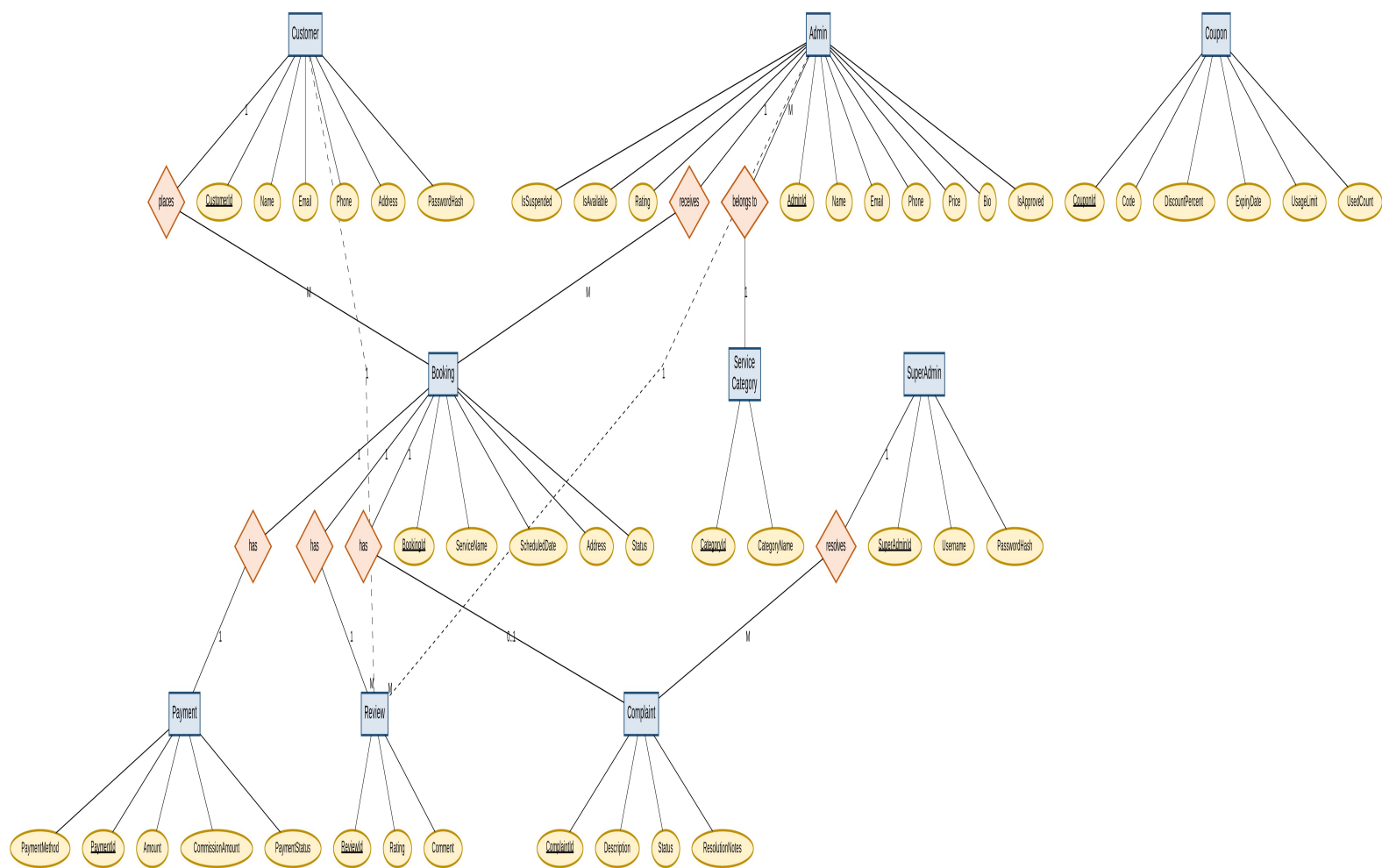


Figure 1: FixHub Entity-Relationship Diagram

Normalization & Finalization

Each one-to-many relationship identified in the Chen ER diagram is normalized below, starting from an Unnormalized Form (UNF) that combines both sides' attributes, through First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF). Since every attribute in FixHub's data model is already atomic (no repeating groups or multi-valued fields), 1NF is identical to UNF in every case below; the meaningful decomposition happens at 2NF, where each relationship is split so that every non-key attribute depends only on its own table's primary key, removing the partial dependency introduced by combining both entities into one row.

Places (Customer → Booking): one-to-many relationship.

UNF: CustomerId, Name, Email, Phone, Address, PasswordHash, BookingId, ServiceName, ScheduledDate, Address, Status.

1NF: CustomerId, Name, Email, Phone, Address, PasswordHash, BookingId, ServiceName, ScheduledDate, Address, Status.

2NF:

1. CustomerId, Name, Email, Phone, Address, PasswordHash.
2. BookingId, ServiceName, ScheduledDate, Address, Status, CustomerId (FK).

3NF: Same as 2NF — no attribute depends on another non-key attribute.

Receives (Admin → Booking): one-to-many relationship.

UNF: AdminId, Name, Email, Phone, Price, Bio, IsApproved, IsSuspended, IsAvailable, Rating, BookingId, ServiceName, ScheduledDate, Address, Status.

1NF: AdminId, Name, Email, Phone, Price, Bio, IsApproved, IsSuspended, IsAvailable, Rating, BookingId, ServiceName, ScheduledDate, Address, Status.

2NF:

1. AdminId, Name, Email, Phone, Price, Bio, IsApproved, IsSuspended, IsAvailable, Rating.
2. BookingId, ServiceName, ScheduledDate, Address, Status, AdminId (FK).

3NF: Same as 2NF.

Belongs To (Admin → ServiceCategory): many-to-one relationship.

UNF: AdminId, Name, Email, Phone, Price, Bio, IsApproved, IsSuspended, IsAvailable, Rating, CategoryId, CategoryName.

1NF: AdminId, Name, Email, Phone, Price, Bio, IsApproved, IsSuspended, IsAvailable, Rating, CategoryId, CategoryName.

2NF:

1. AdminId, Name, Email, Phone, Price, Bio, IsApproved, IsSuspended, IsAvailable, Rating, CategoryId (FK).

2. CategoryId, CategoryName.

3NF: Same as 2NF.

Has — Payment (Booking → Payment): one-to-one relationship.

UNF: BookingId, ServiceName, ScheduledDate, Address, Status, PaymentId, Amount, CommissionAmount, PaymentStatus, PaymentMethod.

1NF: BookingId, ServiceName, ScheduledDate, Address, Status, PaymentId, Amount, CommissionAmount, PaymentStatus, PaymentMethod.

2NF:

1. BookingId, ServiceName, ScheduledDate, Address, Status.
2. PaymentId, Amount, CommissionAmount, PaymentStatus, PaymentMethod, BookingId (FK).

3NF: Same as 2NF.

Has — Review (Booking → Review): one-to-one relationship.

UNF: BookingId, ServiceName, ScheduledDate, Address, Status, ReviewId, Rating, Comment.

1NF: BookingId, ServiceName, ScheduledDate, Address, Status, ReviewId, Rating, Comment.

2NF:

1. BookingId, ServiceName, ScheduledDate, Address, Status.
2. ReviewId, Rating, Comment, BookingId (FK), CustomerId (FK), AdminId (FK).

3NF: Same as 2NF. CustomerId and AdminId are carried onto Review as foreign keys so a review can be read directly without re-joining through Bookings; both are still fully dependent only on ReviewId, so no transitive dependency is introduced.

Has — Complaint (Booking → Complaint): one-to-zero-or-one relationship.

UNF: BookingId, ServiceName, ScheduledDate, Address, Status, ComplaintId, Description, Status, ResolutionNotes.

1NF: BookingId, ServiceName, ScheduledDate, Address, Status, ComplaintId, Description, Status, ResolutionNotes.

2NF:

1. BookingId, ServiceName, ScheduledDate, Address, Status.
2. ComplaintId, Description, Status, ResolutionNotes, BookingId (FK).

3NF: Same as 2NF.

Resolves (SuperAdmin → Complaint): one-to-many relationship.

UNF: SuperAdminId, Username, PasswordHash, ComplaintId, Description, Status, ResolutionNotes.

1NF: SuperAdminId, Username, PasswordHash, ComplaintId, Description, Status, ResolutionNotes.

2NF:

1. SuperAdminId, Username, PasswordHash.
2. ComplaintId, Description, Status, ResolutionNotes, ResolvedBySuperAdminId (FK).

3NF: Same as 2NF.

Finalization

The 2NF/3NF decompositions above are merged wherever the same table was produced by more than one relationship (for example, Booking is produced by both Places and Receives, and Complaint is produced by both Has—Complaint and Resolves), giving the following finalized tables:

1. Customer: CustomerId, Name, Email, Phone, Address, PasswordHash.
2. Admin: AdminId, Name, Email, Phone, Price, Bio, IsApproved, IsSuspended, IsAvailable, Rating, CategoryId (FK).
3. ServiceCategory: CategoryId, CategoryName.
4. Booking: BookingId, ServiceName, ScheduledDate, Address, Status, CustomerId (FK), AdminId (FK).
5. Payment: PaymentId, Amount, CommissionAmount, PaymentStatus, PaymentMethod, BookingId (FK).
6. Review: ReviewId, Rating, Comment, BookingId (FK), CustomerId (FK), AdminId (FK).
7. Complaint: ComplaintId, Description, Status, ResolutionNotes, BookingId (FK), ResolvedBySuperAdminId (FK).
8. SuperAdmin: SuperAdminId, Username, PasswordHash.

Final Tables

Coupon is not produced by any relationship decomposition above, since it has no foreign key linking it to Booking in the current schema (see the note in the ER Diagram section); it is already in 3NF as a standalone table: CouponId, Code, DiscountPercent, ExpiryDate, UsageLimit, UsedCount, IsActive. Together with the eight finalized tables above, this gives the nine tables shown in the Relational Database Schema diagram below.

Relational Database Schema

The Chen ER diagram above translates directly into the following relational schema, showing every table with its columns, data types, primary keys (PK) and foreign keys (FK), as actually implemented in FixHubDb.

Coupons	
CouponId	int, PK
Code	nvarchar(20), UQ
DiscountPercent	decimal(5,2)
ExpiryDate	date
UsageLimit	int
UsedCount	int
IsActive	bit

SuperAdmins	
SuperAdminId	int, PK
Username	nvarchar(50), UQ
PasswordHash	nvarchar(255)
CreatedAt	datetime

Payments	
PaymentId	int, PK
BookingId	int, FK
Amount	decimal(10,2)
CommissionAmount	decimal(10,2)
PaymentStatus	nvarchar(20)
PaymentMethod	nvarchar(20)
CreatedAt	datetime

Complaints	
ComplaintId	int, PK
BookingId	int, FK
Description	nvarchar(max)
Status	nvarchar(20)
ResolutionNotes	nvarchar(max)
ResolvedBySuperAdminId	int
ResolvedAt	datetime
CreatedAt	datetime

Bookings	
BookingId	int, PK
CustomerId	int, FK
AdminId	int, FK
ServiceName	nvarchar(150)
ScheduledDate	datetime
Address	nvarchar(500)
Status	nvarchar(20), CHECK
CreatedAt	datetime

Customers	
CustomerId	int, PK
Name	nvarchar(150)
Email	nvarchar(255), UQ
Phone	nvarchar(20)
Address	nvarchar(500)
PasswordHash	nvarchar(255)
CreatedAt	datetime

Reviews	
ReviewId	int, PK
BookingId	int, FK
CustomerId	int, FK
AdminId	int, FK
Rating	int CHECK 1-5
Comment	nvarchar(500)
CreatedAt	datetime

Admins	
AdminId	int, PK
Name	nvarchar(150)
Email	nvarchar(255), UQ
Phone	nvarchar(20)
CategoryId	int, FK
Price	decimal(10,2)
Bio	nvarchar(max)
PasswordHash	nvarchar(255)
IsApproved	bit
IsSuspended	bit
IsAvailable	bit
Rating	decimal(3,2)
CreatedAt	datetime

ServiceCategories	
CategoryId	int, PK
CategoryName	nvarchar(100), UQ

Figure 2: FixHub Relational Database Schema

SQL Queries

The complete Data Definition Language (DDL) script below creates the FixHubDb database and every table shown in the schema diagram, including primary keys, foreign key constraints, and check constraints. This script is idempotent — each table is only created if it does not already exist — so it is safe to run multiple times against the same database.

Database

```
CREATE DATABASE FixHubDb;  
GO  
USE FixHubDb;  
GO
```

ServiceCategories Table

```
CREATE TABLE ServiceCategories (  
    CategoryId INT PRIMARY KEY IDENTITY(1,1),  
    CategoryName NVARCHAR(100) NOT NULL UNIQUE  
);
```

Customers Table

```
CREATE TABLE Customers (  
    CustomerId INT PRIMARY KEY IDENTITY(1,1),  
    Name NVARCHAR(150) NOT NULL,  
    Email NVARCHAR(255) NOT NULL UNIQUE,  
    Phone NVARCHAR(20) NOT NULL,  
    Address NVARCHAR(500) NOT NULL,  
    PasswordHash NVARCHAR(255) NOT NULL,  
    CreatedAt DATETIME DEFAULT GETDATE()  
);
```

Admins Table (Service Providers)

```
CREATE TABLE Admins (  
    AdminId INT PRIMARY KEY IDENTITY(1,1),  
    Name NVARCHAR(150) NOT NULL,  
    Email NVARCHAR(255) NOT NULL UNIQUE,  
    Phone NVARCHAR(20),  
    CategoryId INT FOREIGN KEY REFERENCES ServiceCategories(CategoryId),  
    Price DECIMAL(10,2) NOT NULL DEFAULT 0,  
    Bio NVARCHAR(MAX),  
    PasswordHash NVARCHAR(255) NOT NULL,  
    IsApproved BIT DEFAULT 0,  
    IsSuspended BIT DEFAULT 0,  
    IsAvailable BIT DEFAULT 1,  
    Rating DECIMAL(3,2) DEFAULT 0,
```

```
    CreatedAt DATETIME DEFAULT GETDATE()  
);
```

Bookings Table

```
CREATE TABLE Bookings (  
    BookingId INT PRIMARY KEY IDENTITY(1,1),  
    CustomerId INT FOREIGN KEY REFERENCES Customers(CustomerId),  
    AdminId INT FOREIGN KEY REFERENCES Admins(AdminId),  
    ServiceName NVARCHAR(150),  
    ScheduledDate DATETIME NOT NULL,  
    Address NVARCHAR(500) NOT NULL,  
    Status NVARCHAR(20) DEFAULT 'Pending',  
    CreatedAt DATETIME DEFAULT GETDATE(),  
    CONSTRAINT CK_Bookings_Status CHECK (Status IN  
        ('Pending', 'Accepted', 'In Progress', 'Completed', 'Cancelled', 'Declined'))  
);
```

Payments Table

```
CREATE TABLE Payments (  
    PaymentId INT PRIMARY KEY IDENTITY(1,1),  
    BookingId INT FOREIGN KEY REFERENCES Bookings(BookingId),  
    Amount DECIMAL(10,2) NOT NULL,  
    CommissionAmount DECIMAL(10,2) NOT NULL,  
    PaymentStatus NVARCHAR(20) DEFAULT 'Unpaid',  
    PaymentMethod NVARCHAR(20),  
    CreatedAt DATETIME DEFAULT GETDATE()  
);
```

Reviews Table

```
CREATE TABLE Reviews (  
    ReviewId INT PRIMARY KEY IDENTITY(1,1),  
    BookingId INT FOREIGN KEY REFERENCES Bookings(BookingId),  
    CustomerId INT FOREIGN KEY REFERENCES Customers(CustomerId),  
    AdminId INT FOREIGN KEY REFERENCES Admins(AdminId),  
    Rating INT NOT NULL CHECK (Rating BETWEEN 1 AND 5),  
    Comment NVARCHAR(500),  
    CreatedAt DATETIME DEFAULT GETDATE()  
);
```

SuperAdmins Table

```
CREATE TABLE SuperAdmins (  
    SuperAdminId INT PRIMARY KEY IDENTITY(1,1),  
    Username NVARCHAR(50) NOT NULL UNIQUE,  
    PasswordHash NVARCHAR(255) NOT NULL,  
    CreatedAt DATETIME DEFAULT GETDATE()  
);
```

```
);
```

Complaints Table

```
CREATE TABLE Complaints (  
    ComplaintId INT PRIMARY KEY IDENTITY(1,1),  
    BookingId INT FOREIGN KEY REFERENCES Bookings(BookingId),  
    Description NVARCHAR(MAX) NOT NULL,  
    Status NVARCHAR(20) DEFAULT 'Open',  
    ResolutionNotes NVARCHAR(MAX),  
    ResolvedBySuperAdminId INT,  
    ResolvedAt DATETIME,  
    CreatedAt DATETIME DEFAULT GETDATE()  
);
```

Coupons Table

```
CREATE TABLE Coupons (  
    CouponId INT PRIMARY KEY IDENTITY(1,1),  
    Code NVARCHAR(20) NOT NULL UNIQUE,  
    DiscountPercent DECIMAL(5,2) NOT NULL,  
    ExpiryDate DATE,  
    UsageLimit INT DEFAULT 1,  
    UsedCount INT DEFAULT 0,  
    IsActive BIT DEFAULT 1  
);
```

Sample Reporting Query (JOIN, GROUP BY, Aggregate Function)

The following query, used by the SuperAdmin Reports screen, demonstrates a multi-table JOIN combined with GROUP BY and the SUM aggregate function to calculate total commission earned per service category from completed bookings.

```
SELECT  
    ISNULL(c.CategoryName, 'Other') AS CategoryName,  
    ISNULL(SUM(p.CommissionAmount), 0) AS TotalCommission  
FROM Bookings b  
JOIN Payments p ON b.BookingId = p.BookingId  
JOIN Admins a ON b.AdminId = a.AdminId  
LEFT JOIN ServiceCategories c ON a.CategoryId = c.CategoryId  
WHERE b.Status = 'Completed'  
GROUP BY c.CategoryName  
ORDER BY TotalCommission DESC;
```

Screenshots & Application Walkthrough

This chapter walks through FixHub screen by screen, for each of the three roles, in the order a real user would click through them. Each diagram below shows the actual running application, with numbered steps and arrows labeled with the action (button or link) that carries the user from one screen to the next.

Customer Workflow

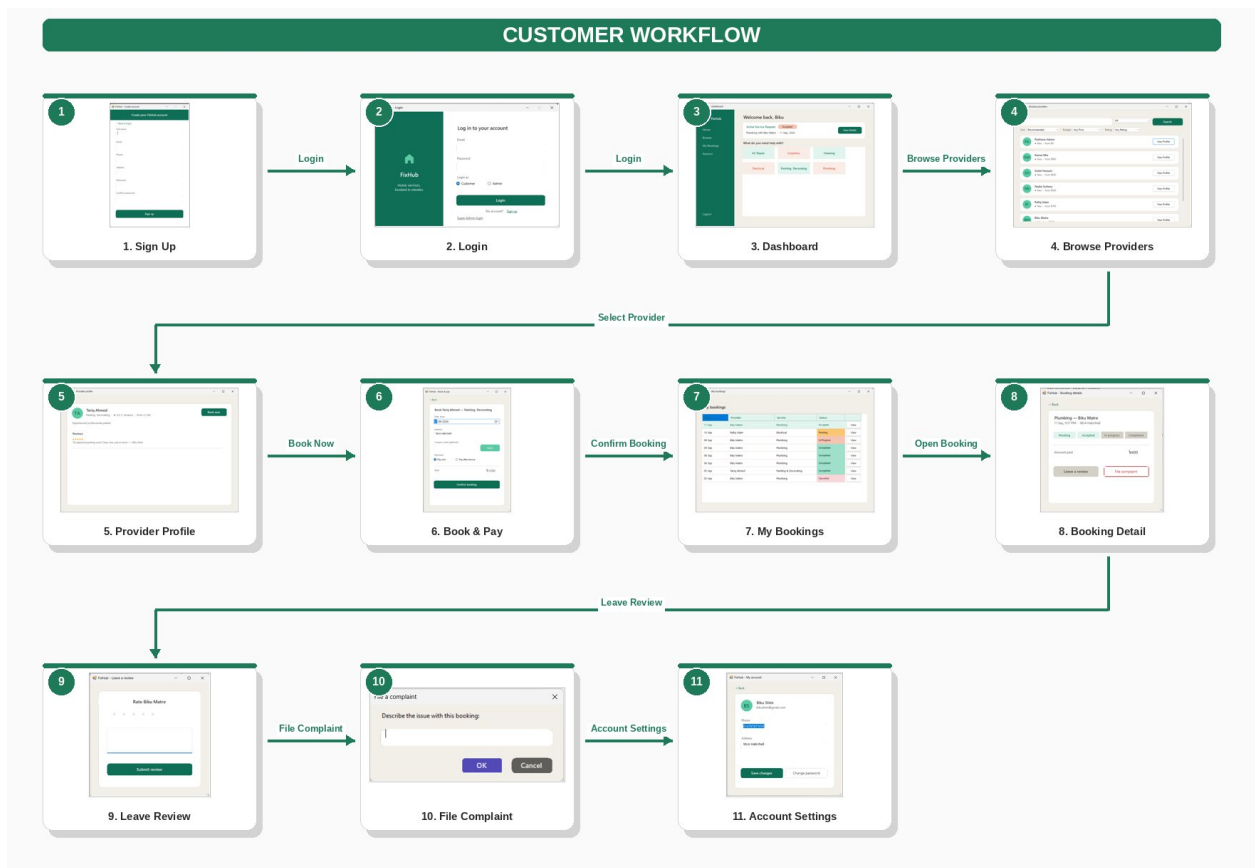


Figure 3: Customer role — screen-by-screen workflow

- 1. Sign Up:** A new customer registers with their name, email, phone, address and password before they can log in.
- 2. Login:** The customer signs in from FixHub's shared entry point, selecting the Customer role.
- 3. Dashboard:** Clicking Login lands the customer on their dashboard, the hub for reaching every other customer screen.
- 4. Browse Providers:** From the dashboard, the customer browses available providers, filterable by category, price range and rating.
- 5. Provider Profile:** Selecting a provider from the list opens their profile: bio, category, price and existing customer reviews.

6. Book & Pay: Clicking Book Now opens the booking form, where the customer confirms the date, address, payment method and can apply a coupon code.

7. My Bookings: After confirming, the customer's booking list shows every booking's provider, date and current status.

8. Booking Detail: Opening a booking from the list shows its full detail, with actions to cancel, leave a review, or file a complaint depending on its status.

9. Leave Review: Once a booking is Completed, the customer can rate the provider from 1 to 5 stars and write a comment.

10. File Complaint: From the same booking detail screen, the customer can instead file a complaint for the SuperAdmin to investigate.

11. Account Settings: Separately, from the dashboard, the customer can update their profile information or change their password.

Admin (Provider) Workflow



Figure 4: Admin (Provider) role — screen-by-screen workflow

1. Provider Sign Up: A service provider registers an Admin account with their category, starting price and bio. The account is created as pending until approved.

2. Admin Dashboard: The Admin logs in through the same shared Login screen as the Customer, selecting the Admin role — FixHub blocks the login with a pending-approval message until the SuperAdmin has approved the account. Once approved, this dashboard is the provider's home.

- 3. Earnings View:** The dashboard's earnings tab shows the provider's total completed bookings and net income after FixHub's commission is deducted.
- 4. Booking Requests:** The Admin views incoming booking requests with the customer, address, scheduled date and status, and accepts, declines or updates them.
- 5. Admin Profile:** The Admin's own profile view shows their bio, category, price and availability toggle.
- 6. Account Update:** The Admin can update their contact details or change their password from here.

SuperAdmin Workflow

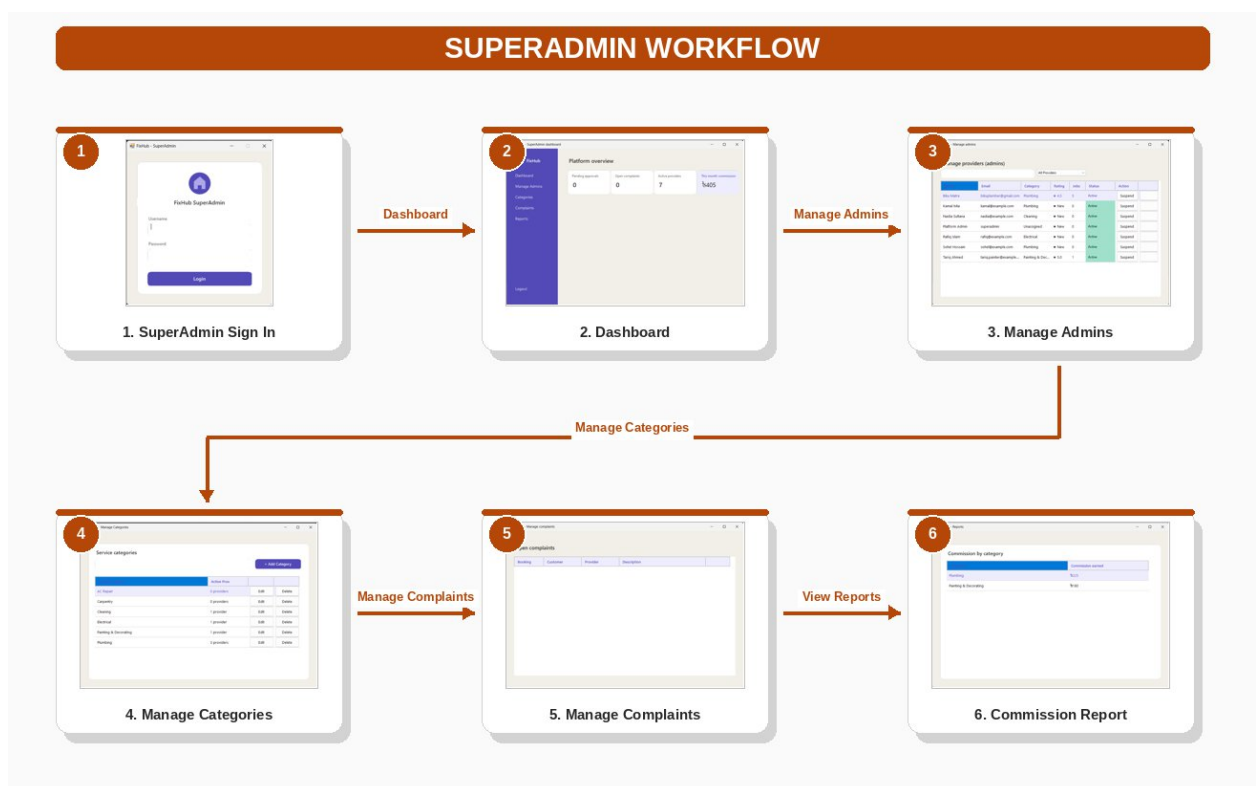


Figure 5: SuperAdmin role — screen-by-screen workflow

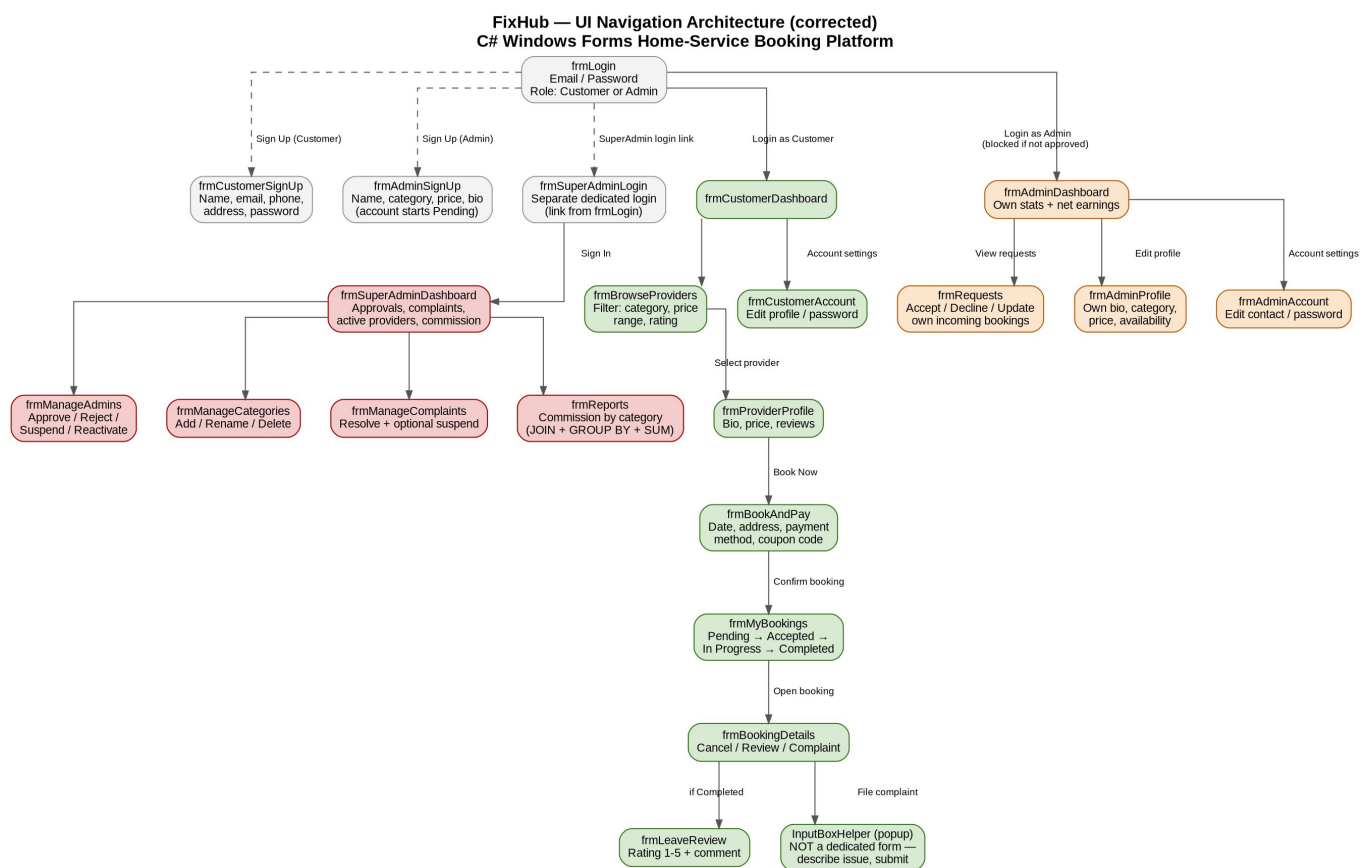
- 1. SuperAdmin Sign In:** The SuperAdmin logs in through a dedicated login form, separate from the Customer/Admin login.
- 2. Dashboard:** Signing in lands the SuperAdmin on a platform-wide dashboard showing pending approvals, open complaints, active providers and total commission earned.
- 3. Manage Admins:** The SuperAdmin reviews providers and approves, rejects, or suspends and later reactivates their accounts.
- 4. Manage Categories:** The SuperAdmin adds, renames or deletes entries in the master service category list used throughout the platform.

5. Manage Complaints: The SuperAdmin reviews open customer complaints and records a resolution, optionally suspending the provider involved.

6. Commission Report: The SuperAdmin views the commission report broken down by service category, generated by joining completed bookings with their payments and the provider's category.

UI Navigation Diagram —

Rebuilt from FixHub's actual 21 form classes (*frmLogin*, *frmBookAndPay*, *frmManageAdmins*, etc.) — replaces the earlier draft, which described forms (*Cart*, *Checkout*, *Invoice*, *Service management*, *commission-percentage settings*) that do not exist in the codebase.



Notation: solid arrow = navigation action · dashed arrow = alternate path (sign up / SuperAdmin login link) · boxes are actual form classes, named to match the source files.

UI

Customer

Form -01: FixHub—Login (Shared)



FixHub
Home services,
booked in minutes

Log in to your account

Email

Password

Login as

☒ Customer ☐ Admin

Login

No account? [Sign up](#)

Form -02: FixHub— SignUp

Create your FixHub account

Full name

Email

Phone

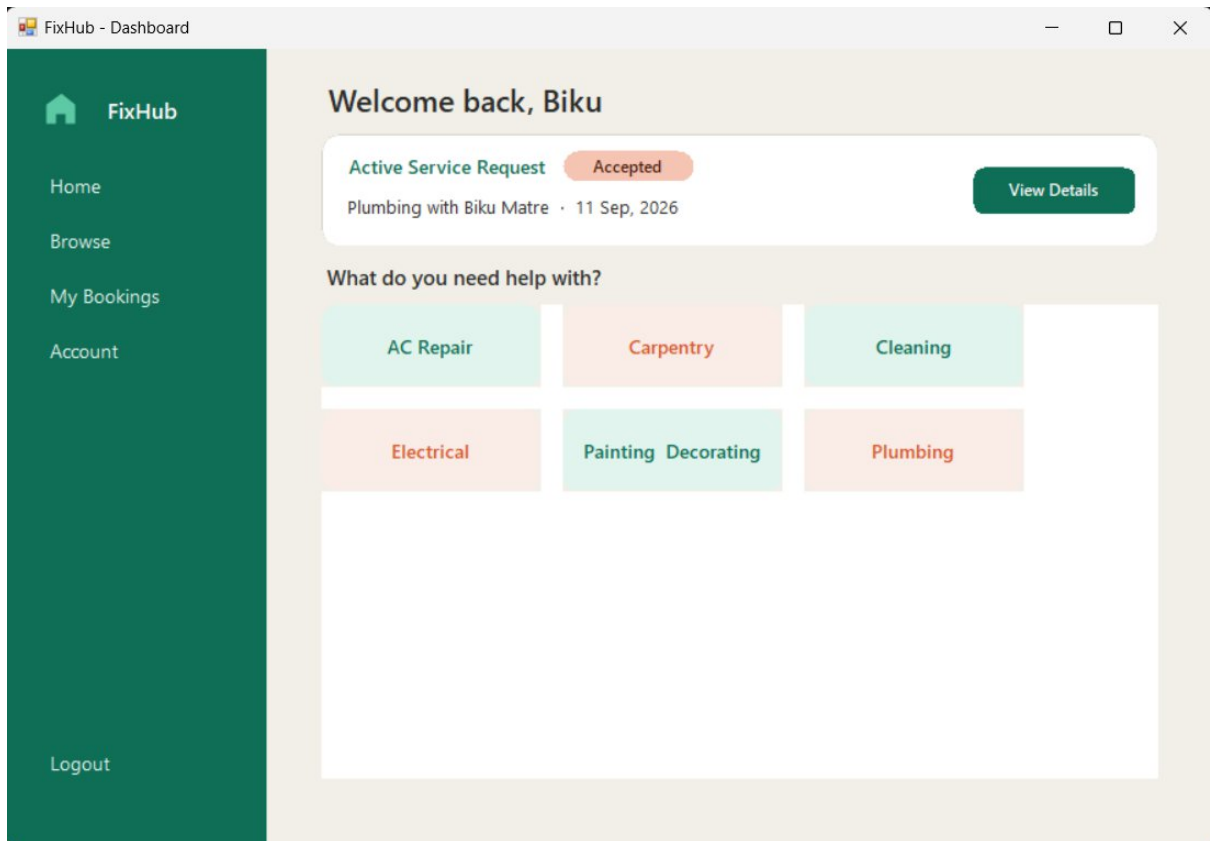
Address

Password

Confirm password

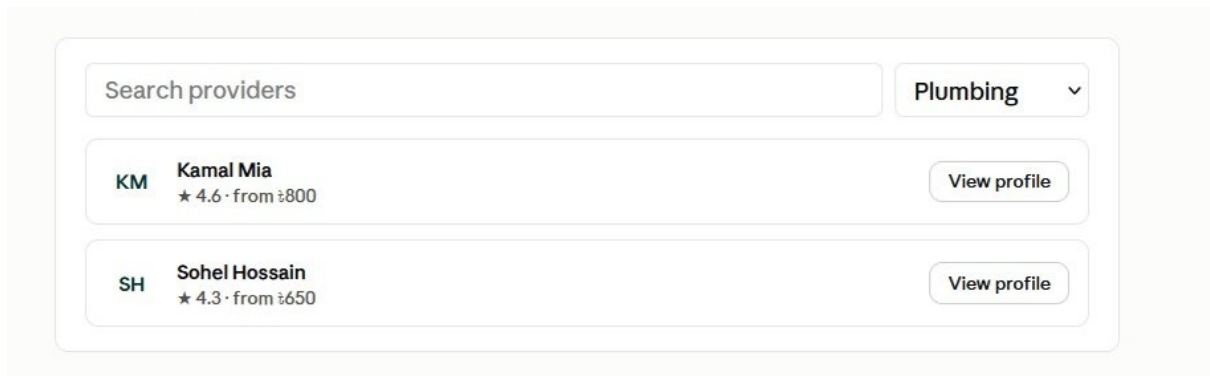
Sign up

Form -03: Customer Dashboard



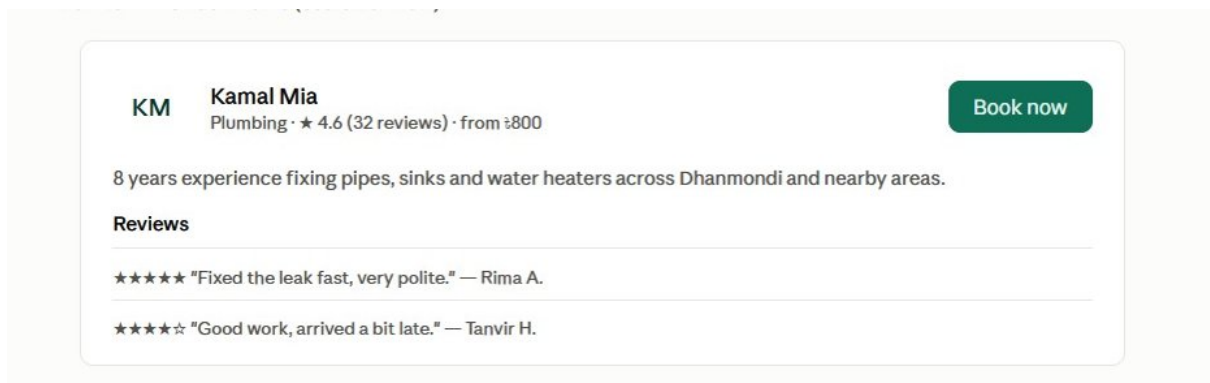
The screenshot shows a web browser window titled "FixHub - Dashboard". On the left is a dark green sidebar with a home icon and the text "FixHub". Below this are links for "Home", "Browse", "My Bookings", "Account", and "Logout". The main content area has a light beige background. At the top, it says "Welcome back, Biku". Below this is a white card with the text "Active Service Request" in green, "Accepted" in an orange pill, and "Plumbing with Biku Matre · 11 Sep, 2026". A green "View Details" button is on the right. Underneath is the heading "What do you need help with?" followed by a grid of six colored buttons: "AC Repair" (teal), "Carpentry" (orange), "Cleaning" (teal), "Electrical" (orange), "Painting Decorating" (teal), and "Plumbing" (orange).

Form -04: Browse Providers



The screenshot shows a search interface for providers. At the top is a search bar labeled "Search providers" and a dropdown menu set to "Plumbing". Below are two provider cards. The first card for "Kamal Mia" (KM) shows a 4.6 star rating and a price of "from ₹800", with a "View profile" button. The second card for "Sohel Hossain" (SH) shows a 4.3 star rating and a price of "from ₹650", also with a "View profile" button.

Form -05: Providers Profile(Customer View)



The screenshot shows a detailed profile for "Kamal Mia" (KM), a plumber with a 4.6 star rating from 32 reviews and a price of "from ₹800". A green "Book now" button is in the top right. Below the name is a bio: "8 years experience fixing pipes, sinks and water heaters across Dhanmondi and nearby areas." The "Reviews" section shows two entries: a 5-star review from Rima A. saying "Fixed the leak fast, very polite." and a 4.5-star review from Tanvir H. saying "Good work, arrived a bit late."

Form -06: Book and Pay

Book Kamal Mia — Plumbing

Date & time

24 Aug, 3:00 PM

Address

House 12, Road 5, Dhanmondi

Coupon code (optional)

FIXFIRST10

Payment

☒ Pay now ☐ Pay after service

Total

₳720

Confirm booking

Form -07: My Bookings

Date	Provider	Status
24 Aug	Kamal Mia	Completed
28 Aug	Sohel Hossain	Pending
02 Sep	Kamal Mia	In progress

Form -08: Booking Details

Sink & pipe repair — Kamal Mia
24 Aug, 3:00 PM · House 12, Road 5, Dhanmondi

Pending

Accepted

In progress

Completed

Amount paid

₳720

Leave a review

File complaint

Form -9: File Complaint

File a complaint

Describe the issue with this booking:

OK

Cancel

Form -10: Customer Review

Rate Kamal Mia



Share a few words about the service

Submit review

Form -11: Customer Account Information

RA

Rima Akter
rima.akter@example.com

Phone

01712345678

Address

House 12, Road 5, Dhanmondi

Save changes

Change password

Admin

Form -01: Admin SignUp(Provider)

Join FixHub as a provider

Full name

Kamal Mia

Email / Phone

kamal@example.com

Service category

Plumbing

Starting price (₹)

800

Bio

Years of experience, specialties...

Password

••••••••

Sign up

Form -02: Admin Dashboard(With earnings Tab)|

FixHub - Admin dashboard

FixHub

Dashboard

Requests

Profile

Account

Logout

Overview

Earnings

Pending

0

This month

₹848

Rating

★4.5

Form -03: Requests(Pending/Accepted In-progress,merged)

Rima Akter Sink repair · 24 Aug 3PM	Accept
Tanvir H. Pipe fitting · 25 Aug 11AM	Start job
Nadia S. Faucet install · 25 Aug 4PM	Mark completed

Form -04: Admin Profile(Editable)

FixHub - My profile

< Back

BM

Biku Matre
★ 4.5 · Plumbing

☒ Available for new bookings

Bio

I am good at plumbing, i can wash it down.

Starting price (₹)

499

Save profile

Form -05: Admin Account/Settings

Account settings

Phone

01798765432

Current password

••••••••

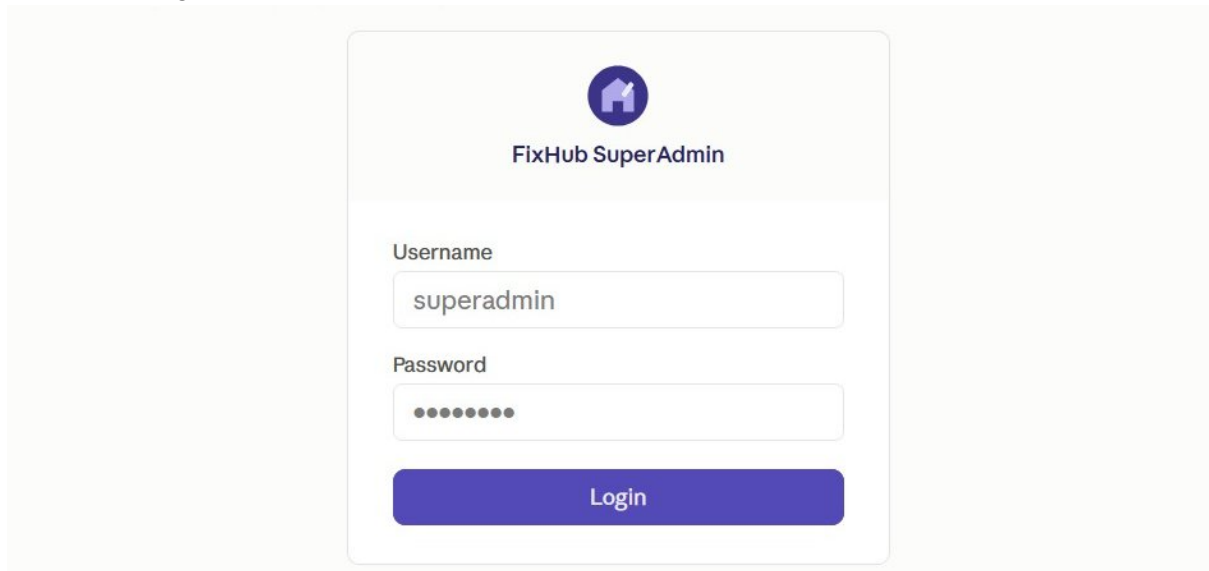
New password

••••••••

Update account

Super Admin

Form -01: Login



The login form is centered on a light gray background. It features a purple circular icon with a white house and a checkmark at the top. Below the icon is the text "FixHub SuperAdmin". The form contains two input fields: "Username" with the value "superadmin" and "Password" with masked characters. A purple "Login" button is at the bottom.

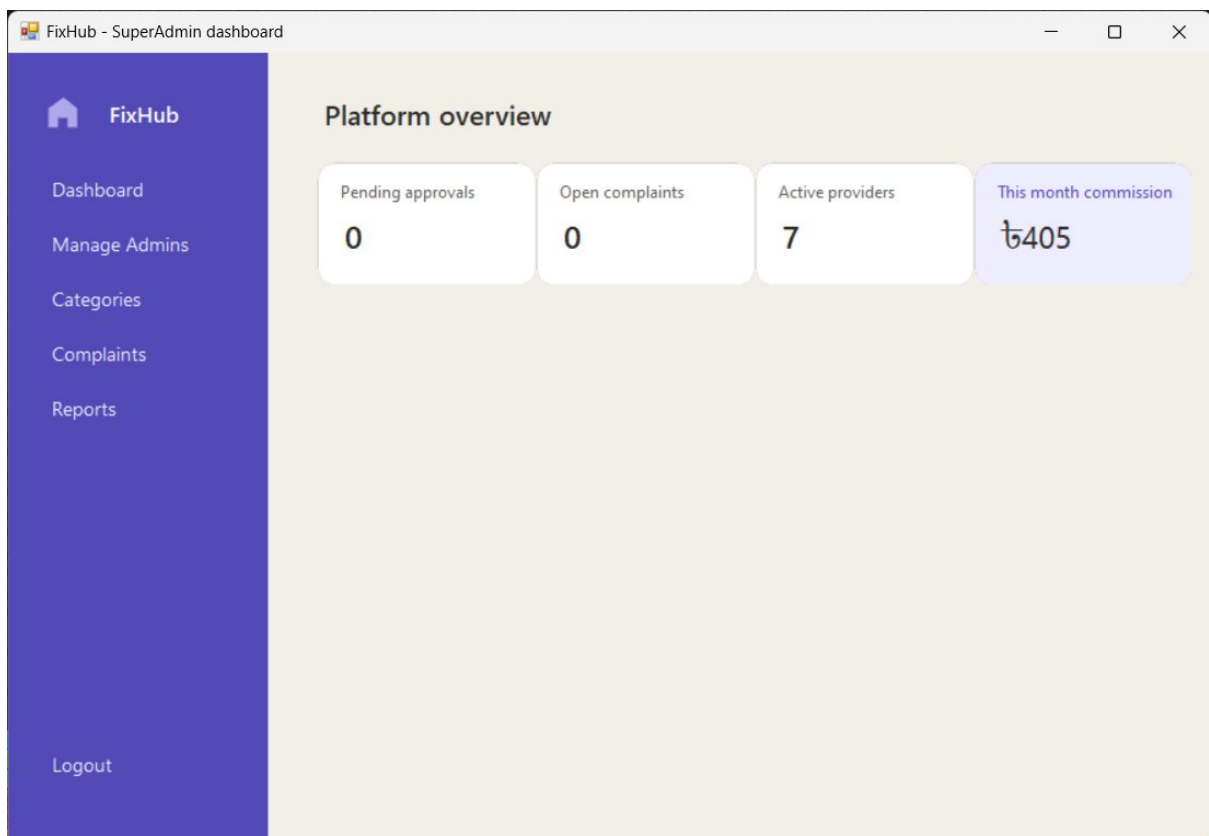
FixHub SuperAdmin

Username
superadmin

Password
●●●●●●●●

Login

Form -02: Dashboard



The dashboard is displayed in a browser window titled "FixHub - SuperAdmin dashboard". It has a purple sidebar with navigation links: "Dashboard", "Manage Admins", "Categories", "Complaints", "Reports", and "Logout". The main content area is titled "Platform overview" and contains four cards: "Pending approvals" (0), "Open complaints" (0), "Active providers" (7), and "This month commission" (₹405).

FixHub - SuperAdmin dashboard

FixHub

Dashboard
Manage Admins
Categories
Complaints
Reports
Logout

Platform overview

Pending approvals	Open complaints	Active providers	This month commission
0	0	7	₹405

Form -03: Manage Complains

Booking #DW3021
Rima A. vs Rafiq Islam · incomplete repair

Resolve

Booking #DW3044
Tanvir H. vs Sohel Hossain · late arrival

Resolve

Form -04: Reports

Commission by category

Plumbing	₹24,100
Electrical	₹18,600
Cleaning	₹12,300

Form 05:- Manage Admins

Kamal Mia
★4.6 · 41 jobs completed

View profile

Sohel Hossain
★4.3 · 27 jobs completed

View profile

Rafiq Islam
★2.1 · 6 complaints

Suspend

Form 06:- Manage Categories

FixHub - Manage Categories

[< Back](#)

Service categories

+ Add Category

Category Name	Active Pros		
AC Repair	0 providers	Edit	Delete
Carpentry	0 providers	Edit	Delete
Cleaning	1 provider	Edit	Delete
Electrical	1 provider	Edit	Delete
Painting & Decorating	1 provider	Edit	Delete
Plumbing	3 providers	Edit	Delete

Work Distribution Table

Name	ID	Percentage
1. Awlad Hossain Abir	24-60107-3	50%
2. Adib Rahman	24-57433-2	25%
3. Sheikh Al Fattah	24-60103-3	25%

Conclusion

FixHub set out to demonstrate a complete, multi-role, database-backed desktop application for CSC 2210's Object-Oriented Programming II final project, and the working system reflects that scope: three separate authentication paths for Customer, Admin and SuperAdmin, a full booking lifecycle from search through payment, review and complaint handling, and a commission-based platform economy with reporting for the platform owner. Choosing a services marketplace over a more common inventory or e-commerce CRUD project pushed the team to model a richer domain: an approval workflow for providers, a booking state machine, and a review system tied to completed transactions rather than to a static product catalogue.

Building FixHub also surfaced real engineering trade-offs rather than only textbook ones. The decision to give each Admin their own profile record instead of modeling a separate "service" entity kept the schema simpler at the cost of a provider only being able to advertise a single price and bio rather than a menu of distinct services. Coupons were implemented as a standalone table validated at booking time rather than as a foreign key on Bookings, which keeps the schema smaller but means the platform cannot currently report which booking used which coupon after the fact. Exception handling throughout the customer and admin forms favours broad, fail-safe catch blocks scoped around individual database calls, so one failed lookup does not take down an entire screen — a pragmatic choice for an academic project, though a production system would narrow these to specific exception types.

For future work, the most valuable next steps would be: recording which coupon was applied to which booking so usage can be audited and reported; allowing a single Admin to list multiple distinct services at different prices rather than one flat rate; adding automated tests around the booking status transitions and commission calculations, which currently have no test coverage; and replacing the simulated Pay Now / Pay After payment flow with a real payment gateway integration. None of these are required for the system to function as demonstrated, but each would move FixHub from a coursework prototype closer to something deployable.